

Third Normal Form (3NF) -- Relational Database Design

Definition: A relation is in Third Normal Form (3NF) if it is in 2NF and no non-prime attribute is transitively dependent on the primary key. For every non-trivial functional dependency $X \rightarrow Y$, either X is a super key or Y is a prime attribute.

1. Detailed Technical Explanation

Transitive Dependency Concept:

A transitive dependency exists if $A \rightarrow B$ and $B \rightarrow C$ hold, which implies $A \rightarrow C$ through non-key attribute B .

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[ Primary Key A ] -----> [ Non-Prime Attribute B ]
                             |
                             v
                        [ Non-Prime Attribute C ]
                    (Transitive Dependency!)
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Conversion Example: 2NF to 3NF

Relation EMP_DEPT(Emp_ID, Emp_Name, Dept_ID, Dept_Name, Dept_Head)

Decomposition into 3NF:

1. EMPLOYEE (Emp_ID, Emp_Name, Dept_ID)

2. Core Concepts & Memory Keywords

- * Transitive Dependency: Non-key attribute determining another non-key attribute.

3. Must-Write Points for Exams

- * 3NF guarantees lossless-join decomposition and dependency preservation.

4. Quick Recall Flow

2NF Relation -> Check $X \rightarrow Y$ -> Ensure X is Super Key OR Y is Prime -> Remove Transitive Dependencies -> 3NF Achieved